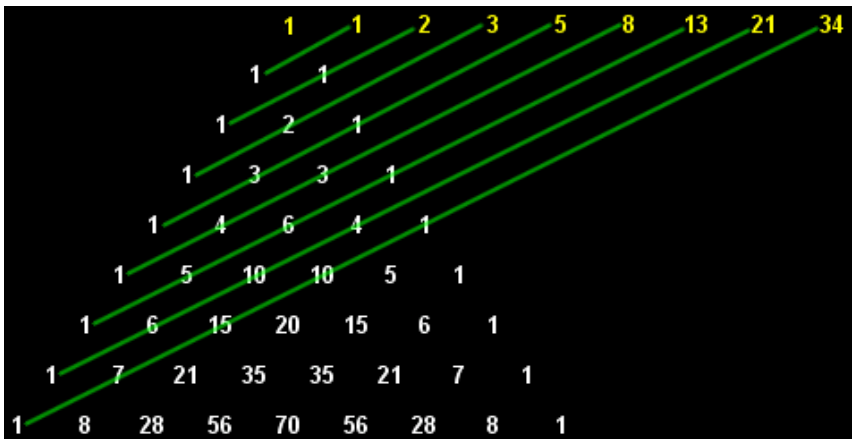
American Computer Science League

2021-2022 ● Contest 3: Fibonacci & Pascal ● Junior Division

PROBLEM: One of the many patterns that can be found in Pascal's Triangle is where the Fibonacci sequence 1, 1, 2, 3, 5, 8, 13, 21, 34, … can be found by adding the numbers on the diagonals as illustrated with the following diagram:



In this program, generate each row of Pascal's Triangle by placing a 1 in the first and last position. For the positions in the middle of the triangle, every number is the sum of the numbers immediately above it. The 0th row contains a single 1. For row N, there are N+1 items.

INPUT: There are 5 lines of data. Each line has a single integer representing one of the Fibonacci numbers in the following sequence: 1 1 2 3 5 8 13 21 34 55 89 .... We guarantee it will be less than 1,000,000.

OUTPUT: For each line of data, print the integers on the diagonal of Pascal's Triangle that add up to the given number, in the order that they appear in the triangle from bottom left to upper right separated by a single space.

SAMPLE INPUT:

```
8
89
610
10946
317811
```

SAMPLE OUTPUT:

```
1. 1 4 3
2. 1 9 28 35 15 1
3. 1 13 66 165 210 126 28 1
4. 1 19 153 680 1820 3003 3003 1716 495 55 1
5. 1 26 300 2024 8855 26334 54264 77520 75582 48620 19448 4368
   455 14
```

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TEST DATA

TEST INPUT:

```
2
55
1597
17711
832040
```

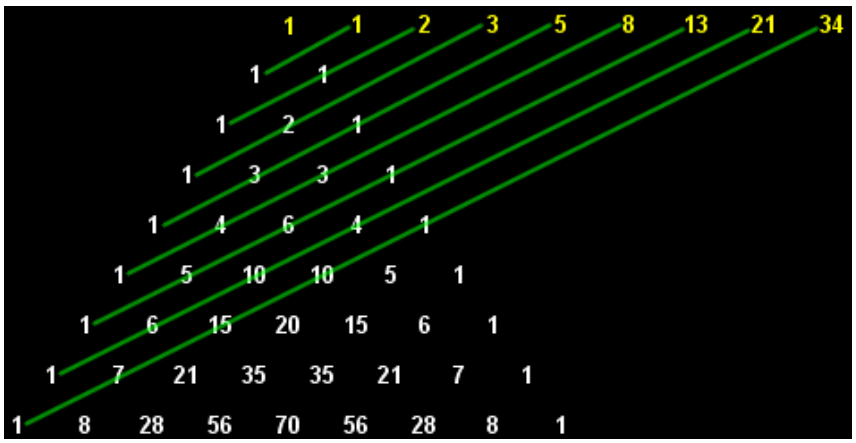
TEST OUTPUT:

1. 1 1
2. 1 8 21 20 5
3. 1 15 91 286 495 462 210 36 1
4. 1 20 171 816 2380 4368 5005 3432 1287 220 11
5. 1 28 351 2600 12650 42504 100947 170544 203490 167960 92378 31824 6188 560 15

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题目：在帕斯卡三角形**中能够发现一些性质，如通过将帕斯卡三角形对角线上的数字相加，便可以找到斐波那契数列** 1, 1, 2, 3, 5, 8, 13, 21, 34, …，**如下图所示：**



在上图中，**帕斯卡三角形的每一行的首尾位置都是** 1 。**在三角形中间位置上的每个数字**都是**其上方紧邻的两个数字之和** 。**第** 0 **行**只有**一个数字** 1 。 **第** N 行**有** N+1 **个数字。**

输入：输入 5 行**数据。每行都**有**一个整数，表示斐波那契数列中的一个元素：** 1 1 2 3 5 8 13 21 34 55 89 .... 我们保证这个整数**不会超过** 1,000,000。

输出：根据输入的每行数据，输出帕斯卡三角形中某条对角线上的所有数字，这些数字相加之和等于给定整数，按照从**三角形左下方到右上方**出现的顺序**依次输出，中间由一个空格隔开。**

示例输入：

```
8
89
610
10946
317811
```

示例输出:

```
1. 1 4 3
2. 1 9 28 35 15 1
3. 1 13 66 165 210 126 28 1
4. 1 19 153 680 1820 3003 3003 1716 495 55 1
5. 1 26 300 2024 8855 26334 54264 77520 75582 48620 19448 4368
   455 14
```

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测试数据

测试输入：

```
2
55
1597
17711
832040
```

测试输出：

```
1. 1 1
2. 1 8 21 20 5
3. 1 15 91 286 495 462 210 36 1
4. 1 20 171 816 2380 4368 5005 3432 1287 220 11
5. 1 28 351 2600 12650 42504 100947 170544 203490 167960 92378
   31824 6188 560 15
```